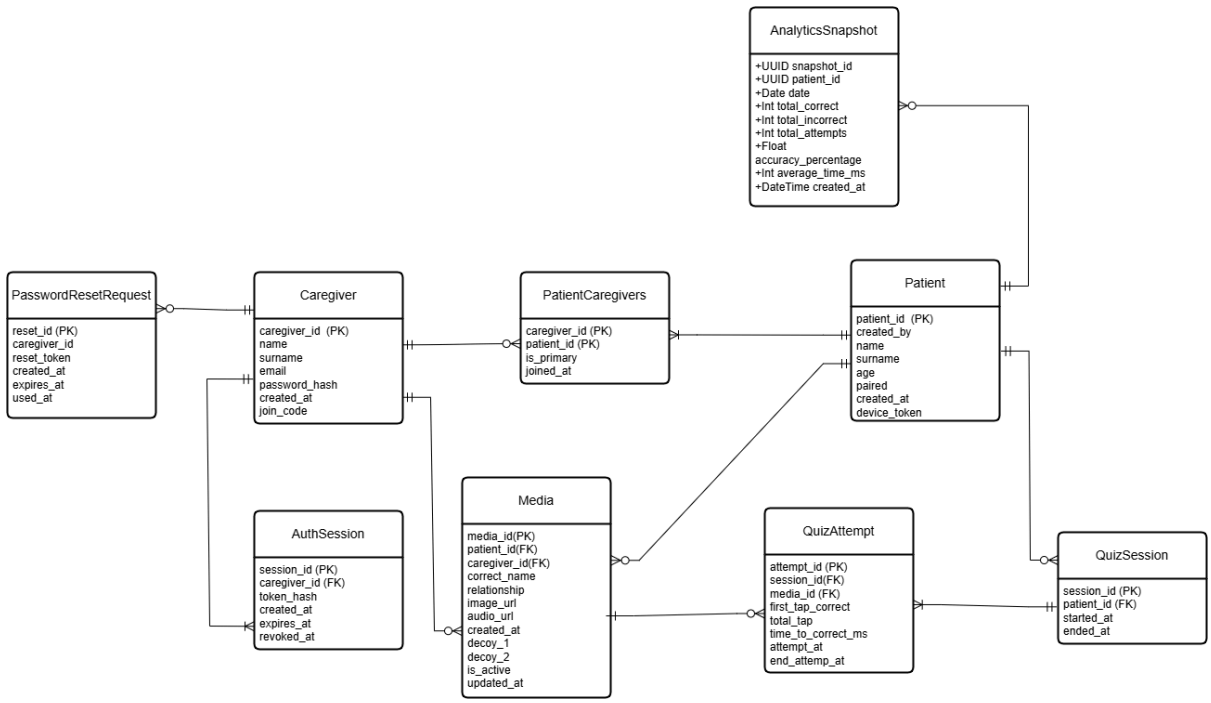
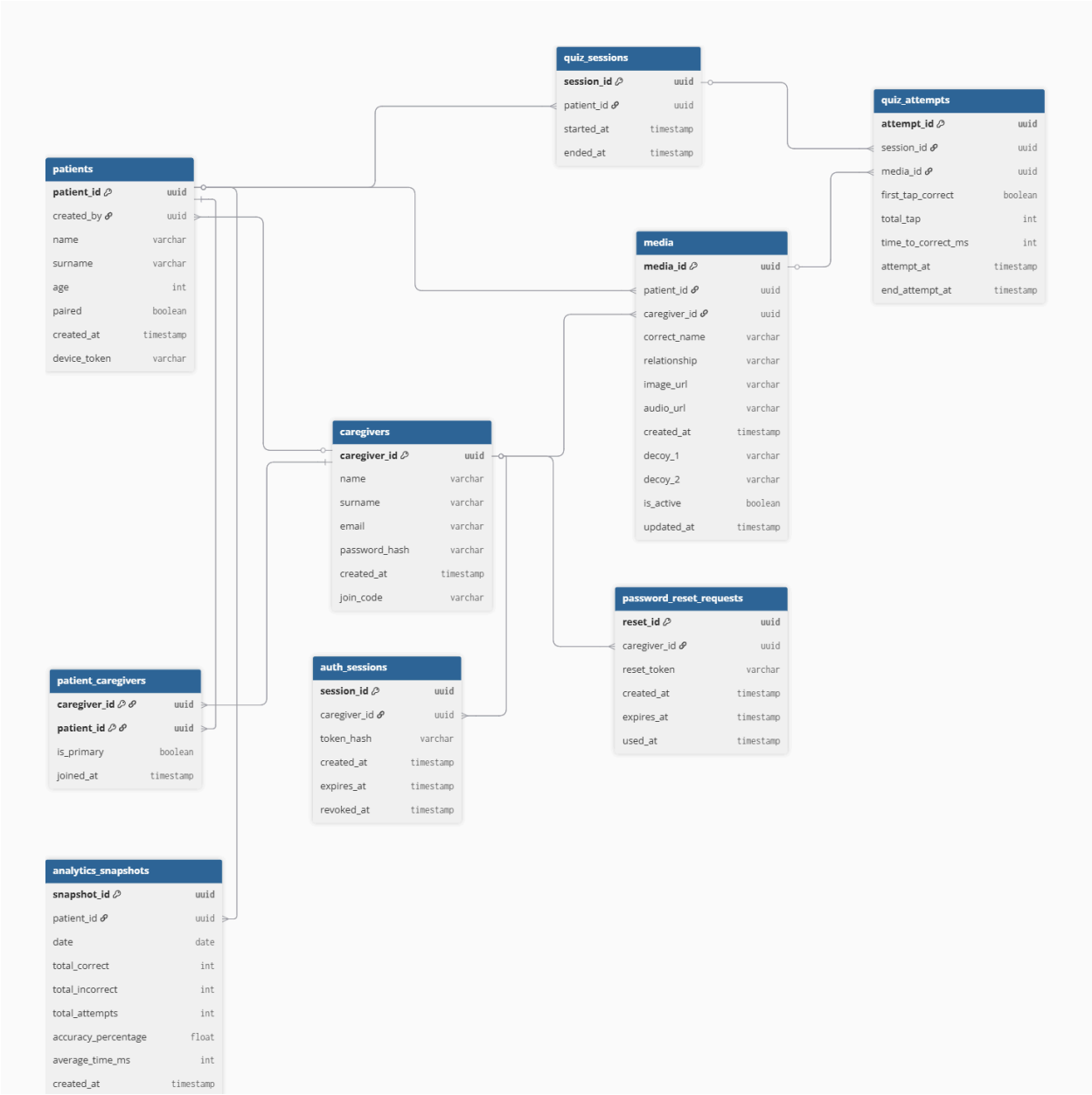
MemoryLane - System Design

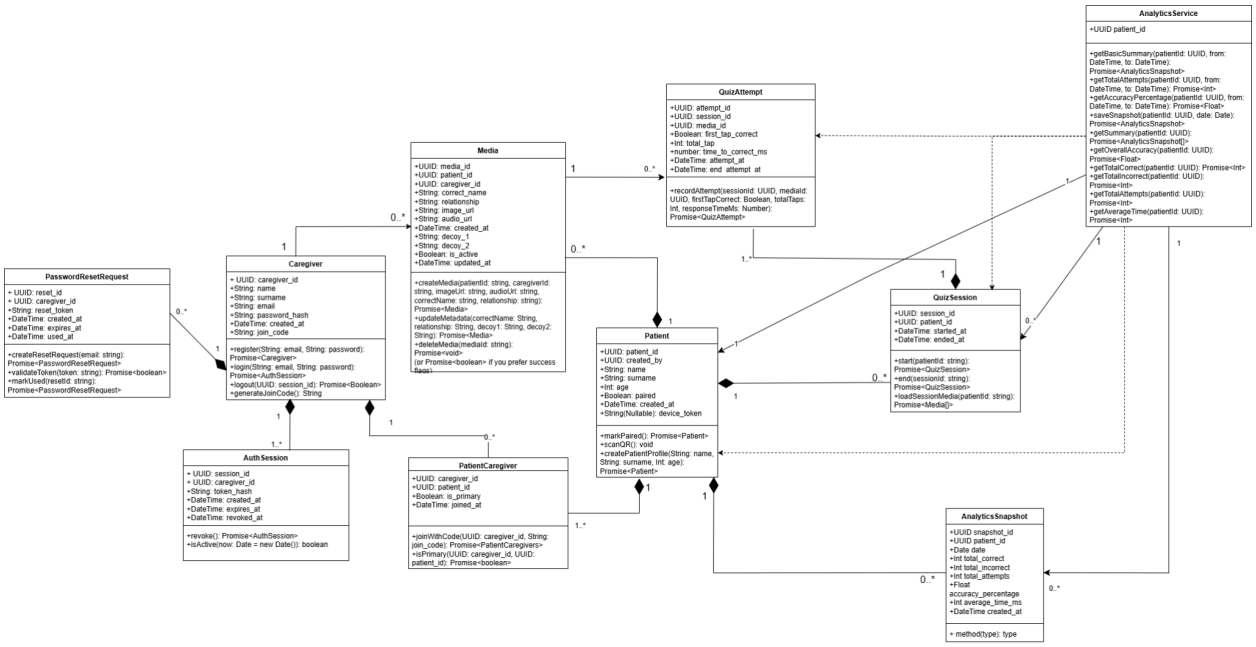
Entity-Relationship Diagram



Relational-Schema



Class Diagram



Prisma Schema

```
generator client {
provider = "prisma-client-js"
        }

datasource db {
provider = "postgresql"
url      = env("DATABASE_URL")
}

model Caregiver {
id                    String                    @id @default(uuid())
name                  String
surname               String
email                 String                    @unique
passwordHash          String                    @map("password_hash")
createdAt             DateTime                  @default(now())
@map("created_at")
joinCode              String                    @unique @map("join_code")

patientCaregivers     PatientCaregiver[]
uploadedMedia         Media[]
authSessions          AuthSession[]
passwordResetRequests PasswordResetRequest[]
createdPatients       Patient[]
```

```
@@map("caregivers")
}

model Patient {
id                  String              @id @default(uuid())
createdBy           String              @map("created_by")
name                String
surname             String
age                 Int?
joinCode            String              @unique @map("join_code")
networkCode         String              @unique @map("network_code")
paired              Boolean             @default(false)
createdAt           DateTime            @default(now()) @map("created_at")
deviceToken         String?             @map("device_token")

creator             Caregiver           @relation(fields: [createdBy],
references: [id])
patientCaregivers   PatientCaregiver[]
media               Media[]
quizSessions        QuizSession[]
analyticsSnapshots  AnalyticsSnapshot[]

@@map("patients")
}

model PatientCaregiver {
caregiverId String    @map("caregiver_id")
patientId   String    @map("patient_id")
isPrimary   Boolean   @default(false) @map("is_primary")
joinedAt    DateTime  @default(now()) @map("joined_at")

caregiver   Caregiver @relation(fields: [caregiverId], references: [id])
patient     Patient   @relation(fields: [patientId], references: [id])

@@id([caregiverId, patientId])
@@map("patient_caregivers")
}

model Media {
id            String        @id @default(uuid())
patientId     String        @map("patient_id")
caregiverId   String?       @map("caregiver_id")
correctName   String        @map("correct_name")
relationship  String
imageUrl      String        @map("image_url")
audioUrl      String        @map("audio_url")
decoy1        String        @map("decoy_1")
```

```
decoy2       String        @map("decoy_2")
isActive     Boolean       @default(true) @map("is_active")
createdAt    DateTime      @default(now()) @map("created_at")
updatedAt    DateTime      @updatedAt @map("updated_at")

patient      Patient       @relation(fields: [patientId], references:
[id])
caregiver    Caregiver?    @relation(fields: [caregiverId], references:
[id])
quizAttempts QuizAttempt[]

@@map("media")
}

model QuizSession {
id           String        @id @default(uuid())
patientId    String        @map("patient_id")
startedAt    DateTime      @default(now()) @map("started_at")
endedAt      DateTime?     @map("ended_at")

patient      Patient       @relation(fields: [patientId], references:
[id])
quizAttempts QuizAttempt[]

@@map("quiz_sessions")
}

model QuizAttempt {
id              String      @id @default(uuid())
sessionId       String      @map("session_id")
mediaId         String      @map("media_id")
firstTapCorrect Boolean     @map("first_tap_correct")
totalTaps       Int         @map("total_taps")
timeToCorrectMs Int         @map("time_to_correct_ms")
attemptedAt     DateTime    @default(now()) @map("attempted_at")
endAttemptAt    DateTime?   @map("end_attempt_at")

session         QuizSession @relation(fields: [sessionId], references:
[id])
media           Media       @relation(fields: [mediaId], references: [id])

@@map("quiz_attempts")
}

model AuthSession {
id          String    @id @default(uuid())
caregiverId String    @map("caregiver_id")
```

```
tokenHash   String    @map("token_hash")
createdAt   DateTime  @default(now()) @map("created_at")
expiresAt   DateTime  @map("expires_at")
revokedAt   DateTime? @map("revoked_at")

caregiver   Caregiver @relation(fields: [caregiverId], references: [id])

@@map("auth_sessions")
}

model PasswordResetRequest {
id          String    @id @default(uuid())
caregiverId String    @map("caregiver_id")
resetToken  String    @map("reset_token")
createdAt   DateTime  @default(now()) @map("created_at")
expiresAt   DateTime  @map("expires_at")
usedAt      DateTime? @map("used_at")

caregiver   Caregiver @relation(fields: [caregiverId], references: [id])

@@map("password_reset_requests")
}

model AnalyticsSnapshot {
id                  String   @id @default(uuid())
patientId           String   @map("patient_id")
date                DateTime @db.Date
totalCorrect        Int      @map("total_correct")
totalIncorrect      Int      @map("total_incorrect")
totalAttempts       Int      @map("total_attempts")
accuracyPercentage  Float    @map("accuracy_percentage")
averageTimeMs       Int      @map("average_time_ms")
createdAt           DateTime @default(now()) @map("created_at")

patient             Patient  @relation(fields: [patientId], references:
[id])

@@map("analytics_snapshots")
}
```